

Back to Basics: Fabrication of Parts



Agenda

- Review of the regulations and guidance regarding the fabrication of parts consumed during maintenance and alterations.



Reference:

- 14 CFR § 1.1 General definitions.
- 14 CFR § 3.5 Statements about products, parts, appliances and materials.
- 14 CFR § 21.8 Approval of articles.
- 14 CFR § 21.9 Replacement and modification articles.
- 14 CFR § 21.31 Type design.
- 14 CFR § 43.13 Performance rules (general)
- AC 43-18: Fabrication of Aircraft Parts by Maintenance Personnel



14 CFR § 1.1 General definitions.

- Maintenance means inspection, overhaul, repair, preservation, and the replacement of parts, but excludes preventive maintenance.



14 CFR § 18.4 Alteration

- An alteration is any appreciable change in the design of an aircraft.
Aircraft engine, propeller, or instrument.



Three requirements for Maintenance and Alterations:

- Qualified and certificated person.
- Parts “pedigree”
- Installation data.



Three requirements for Maintenance and Alterations:

- Certificated and Qualified person.
- **Parts “pedigree”**
- Installation data.



14 CFR § 3.5 Statements about products, parts, appliances and materials.

- (a) Definitions.
- Airworthy means the aircraft conforms to its type design and is in a condition for safe operation.



Regulations for parts

- 14 CFR § 21.8 Approval of articles.
- 14 CFR § 21.9 Replacement and modification articles.
- 14 CFR § 43.13 Performance rules (general).



14 CFR § 43.13 Performance rules (general).

- (b) Each person maintaining or altering, or performing preventive maintenance, shall do that work in such a manner and use materials of such a quality, that the condition of the aircraft, airframe, aircraft engine, propeller, or appliance worked on will be at least equal to its original or properly altered condition (with regard to aerodynamic function, structural strength, resistance to vibration and deterioration, and other qualities affecting airworthiness).



14 CFR § 21.8 Approval of articles.

If an article is required to be approved under this chapter, it may be approved—

- (a) Under a PMA;
- (b) Under a TSO;
- (c) In conjunction with type certification procedures for a product; or
- (d) In any other manner approved by the FAA.



14 CFR § 21.9 Replacement and modification articles.

- (a) If a person knows, or should know, that a replacement or modification article is reasonably likely to be installed on a type-certificated product, the person may not produce that article unless it is -



14 CFR § 21.9 Replacement and modification articles.

(a) ... the person may not produce that article unless it is—

- (1) Produced under a type certificate;
- (2) Produced under an FAA production approval;



14 CFR § 21.9 Replacement and modification articles.

(a) ... the person may not produce that article unless it is—

(3) A standard part (such as a nut or bolt) manufactured in compliance with a government or established industry specification;

(4) A commercial part as defined in § 21.1 of this part;



14 CFR § 21.9 Replacement and modification articles.

(a) ... the person may not produce that article unless it is—

(5) Produced by an owner or operator for maintaining or altering that owner or operator's product;

(6) *Fabricated* by an appropriately rated certificate holder with a quality system, *and consumed in the repair or alteration* of a product or article in accordance with part 43 of this chapter; or



Subcontractor Limitations

- When a subcontractor is used in the fabrication process, the certificate holder under whose surveillance the fabrication occurs must control the design, manufacture, and quality of the part.



Subcontractor Limitations

- The work performed by the subcontractor must be documented to support a determination of conformance to the purchase order requirements and substantiated by a maintenance record.
- The documentation must describe any special processes.



Subcontractor Limitations

- Subcontractors must be subject to control and audit by the certificate holder fabricating the part for return to service.



AC 43-18, 3 (b): Airworthy.

The term “airworthy” is defined in 14 CFR part 3, § 3.5(a); a clear understanding of its meaning is essential in making an airworthiness determination.

The definition of airworthy applies to type-certificated products (aircraft, aircraft engine, or propeller), *and parts thereof*. Title 49 of the United States Code (49 U.S.C.) § 44704(c) and 14 CFR part 21, § 21.183(a), (b), and (c) state that the two conditions that must be met for issuance of an airworthiness certificate are:



AC 43-18, 3 (b): Airworthy.

.... two conditions that must be met for issuance of an airworthiness certificate are:

(1) The product must conform to its type certificate (TC).

A product conforms to its TC when its configuration and the components installed are as described in the drawings, specifications, and other data that are part of the TC, which includes any Supplemental Type Certificates (STC), Airworthiness Directives (AD), and field-approved alterations incorporated into the product; and



AC 43-18, 3 (b): Airworthy.

.... two conditions that must be met for issuance of an airworthiness certificate are:

(2) The aircraft (product) must be in a condition for safe operation.

NOTE:

If one or more of these conditions are not satisfied, the product would not be considered airworthy.



14 CFR § 21.31 Type design.

The type design consists of—

(a) The drawings and specifications, and a listing of those drawings and specifications, necessary to define the configuration and the design features of the product shown to comply with the requirements of that part of this subchapter applicable to the product;



14 CFR § 21.31 Type design.

The type design consists of—

(b) Information on dimensions, materials, and processes necessary to define the structural strength of the product;



14 CFR § 21.31 Type design.

The type design consists of—

(c) The Airworthiness Limitations section of the Instructions for Continued Airworthiness as required by parts 23, 25, 26, 27, 29, 31, 33 and 35 of this subchapter, or as otherwise required by the FAA; and as specified in the applicable airworthiness criteria for special classes of aircraft defined in § 21.17(b); and



14 CFR § 21.31 Type design.

The type design consists of—

(d) For primary category aircraft, if desired, a special inspection and preventive maintenance program designed to be accomplished by an appropriately rated and trained pilot-owner.



14 CFR § 21.31 Type design.

The type design consists of—

(e) Any other data necessary to allow, by comparison, the determination of the airworthiness, noise characteristics, fuel efficiency, fuel venting, and exhaust emissions (where applicable) of later products of the same type.



AC 43-18, 3 (g): Consumed.

- A fabricated part is considered consumed in a repair when it is installed into the next higher assembly component part, or within a product by the fabricator, while undergoing maintenance or an alteration.



AC 43-18, 3 (i): Fabrication.

- An act in which a part/subpart is made (fabricated) and consumed by the fabricator on the product, or part thereof, in the course of performing maintenance or alterations in accordance with approved or acceptable data, depending on the category (CAT) classification of the part being fabricated and the applicable regulations.



AC 43-18: 6. d. (1) Required Documentation.

- Procedures for addressing the criteria recommended in paragraph 6d(1)(a), 1 through 7 below must be documented and recorded in a manual or similar type of document, easily understood, and readily available to the person(s) fabricating the part(s).
- This data must include the following, dependent on the CAT of the part.
 - (a) For CAT 1 and CAT 2 Parts:
 - (b) For CAT 3 Parts:



CAT 1 Part.

- A fabricated part, the failure of which could prevent continued safe flight and landing; resulting consequences could reduce safety margins, degrade performance, or cause loss of capability to conduct certain flight operations.



CAT 1 Part - Design Issues.

- A CAT 1 part is a part intended to be consumed within a major repair or major alteration. The certificating ACO, through the geographic ACO, must approve the design data.
- The ACO will make the determination of necessary data for development and submittal based on each circumstance. In the case of a CAT 1 part, a DER may only “recommend approval” of the design data.



CAT 2 Part.

- A fabricated part, the failure of which would not prevent continued safe flight and landing, but would reduce the capability of the aircraft or the ability of the flightcrew to cope with adverse operating conditions or subsequent failures.



CAT 2 Part - Design Issues.

- A CAT 2 part is a part intended to be consumed within a major repair or major alteration. Design data is required to be approved by the geographic ACO or appropriately authorized DER.



CAT 3 Part.

A fabricated part, the failure of which would have no effect on the continued safe flight and landing of the aircraft.



CAT 3 Part - Design Issues.

The fabrication of a CAT 3 part will generally require only acceptable data. Fabrication of this type of part will typically result in no involvement by AIR unless the AFS aviation safety inspector (ASI) requests assistance.



Fabrication Quality Control System (FQCS)

Quality control system procedures should be established for ensuring all processes and requirements necessary to fabricate the part are identified and adhered to.



Fabrication Quality Control System (FQCS)

The depth and detail of the FQCS depends on the complexity and CAT classification of the part being fabricated.



Part Marking.

Fabricated parts must be clearly identified with an additional permanent and legible marking.

(a) The marking must include the following:

1. The name, trademark, or symbol of the FAA certificate holder (fabricator) under whose control the fabrication occurred;
2. A unique part number that clearly distinguishes the fabricated part; and
3. The original manufacturer's part number if removed as a result of the fabrication.



Part Marking.

(b) Critical parts must be marked in accordance with part 45, § 45.15. This part marking provides traceability for subsequent operators and maintenance providers to the source of the fabricated part.



Part Marking.

NOTE:

In cases where it is impractical to mark the fabricated part without compromising the airworthiness (integrity) of the part, the marking information should be included in the maintenance records for the part or part thereof.



Instructions for Continued Airworthiness (ICA)

Under § 21.50, design approval holders are required to develop and distribute information essential to continued airworthiness of their parts and/or products.

Typically, these instructions are included in maintenance and overhaul manuals to describe the methods, techniques, and practices for performing inspections, maintenance, preventive maintenance, and alterations to ensure that the affected products are maintained in an airworthy condition.



Instructions for Continued Airworthiness (ICA)

Certain sections of the ICA, and any changes to those sections, (e.g., airworthiness limitations, wiring diagrams, or SRM revisions) require FAA approval.

When parts are fabricated during the course of performing maintenance, the fabricator must address the ICA requirements per AC 43-18 (6)(d)(4)



Recordkeeping.

Fabrication of a part in accordance with this AC is subject to the maintenance recordkeeping provisions of § 43.9.

A recordkeeping system should be established for documenting fabricated parts containing sufficient information to determine the airworthiness status of the part.



Recordkeeping.

Information contained in the maintenance record entry could include a description of work performed, date of work completion, person approving return to service, current status of any AD, current inspection status, and current status of life-limited parts.

These maintenance records should be retained with the aircraft records.



Take Away

- Maintenance means inspection, overhaul, repair, preservation, and the replacement of parts, but excludes preventive maintenance.
- Fabrication is an act in which a part/subpart is produced (fabricated) and consumed by the fabricator on the product, or part thereof, in the course of performing maintenance or alterations in accordance with approved or acceptable data.



Take Away

- Fabrication is authorized by 14 CFR § 21.9 Replacement and modification articles.
 - (5) Produced by an owner or operator for maintaining or altering that owner or operator's product.
 - (6) Fabricated by an appropriately rated certificate holder with a quality system, and consumed in the repair or alteration of a product or article in accordance with part 43 of this chapter.



Custom Fabrication Work Request

COMPANY NAME	COMPANY ADDRESS	TELEPHONE: FACSIMILE: W: E:
---------------------	------------------------	--

CUSTOM FABRICATION WORK REQUEST

Fabricator Information

(1) Name _____

(2) Address _____

(3) Telephone _____ Telephone _____

(4) Email _____ Email _____

Fabricator is requested to produce the article identified below under [14](#) CFR § 21.9(a)(5). The fabrication will comply with the Federal Aviation Administration's Chief Counsel opinion referenced in the final rule creating the exception for maintenance fabricated articles, Advisory Circular [43-18](#), and [14](#) CFR part [43](#). The repair station is involved in the design, production, and quality of the articles by providing the design and quality requirements.

Part Description:

(5) Part descriptor _____

(6) Part number _____

Design Requirements

(7) Drawing/Template _____

(8) Materials _____

(9) Part (hardware) _____

standard/specification _____

(10) Dimensions _____

(11) Processes _____

(12) AC 43.13 _____

(13) Other _____

Quality Document Requirements

(14) Fabrication routers/travelers _____

(15) Inspection or laboratory results _____

(16) Certificates of conformity _____

(17) Notes _____



Questions?

